Faruk Pasic - Curriculum Vitae

Date of birth:	6th December 1994
Status:	University Assistant, PhD
Languages:	English, German, Bosnian
Nationality:	Bosnian (Daueraufenthalt - EU)
Address:	Bloch-Bauer-Promenade 10/1/22, 1100 Vienna
Phone:	+43 (676) 7639604
Email:	faruk.pasic.94@gmail.com
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ORCID:	0000-0002-4380-563X
Skills:	Python, MATLAB, C, Git, Microsoft Office, LaTeX, HFSS, Altium, SDR, RF equipment (SA, VNA)

Work experience

04/2026-present

Postdoctoral Researcher — *Vienna, Austria*

Institute of Telecommunications, TU Wien

- Leading collaborative research projects with industry partners on the development and validation of machine-learning-based channel models for next-generation wireless communication systems
- Conducting research in wireless communications and multi-band RF transceiver systems across FR1, FR2 and FR3, with a focus on antenna design, channel measurements, channel modeling, signal processing and system-level performance analysis
- Supervising and mentoring undergraduate students in wireless communications research, measurement campaigns and data analysis

08/2021-03/2026

University Assistant — *Vienna, Austria*

Institute of Telecommunications, TU Wien

- Conducting research in wireless communications with a focus on the design, implementation and experimental characterization of multi-band wireless systems, including sub-6 GHz (FR1) and millimeter-wave (FR2) technologies
- Collaborating with industry partners and external researchers on joint projects
- Contributing to scientific conferences
- Participating actively in the COST Action INTERACT network

04/2024-06/2024

Visiting Researcher — *Lund, Sweden*

Department of Electrical and Information Technology, Lund University

- Conducting dynamic wireless channel measurements using multiple-antenna systems across sub-6 GHz (FR1) and millimeter-wave (FR2) frequency bands
- Enhancing millimeter-wave (FR2) receiver performance through the development and application of machine learning algorithms

06/2024-09/2024

Visiting Researcher — *Los Angeles, CA, USA*

Wireless Devices and Systems Group, University of Southern California

- Developing advanced receiver architectures and signal processing techniques for multi-antenna millimeter-wave (FR2) systems
- Prototyping and evaluating a beamforming wireless system operating in the THz frequency bands

10/2019-07/2021 **Project Assistant** *Vienna, Austria*

Institute of Telecommunications, TU Wien

- Developing and testing a millimeter-wave (FR2) transceiver architectures
- Designing and prototyping RF front-ends on printed circuit boards (PCBs)
- Antenna design and characterization

Education

09/2021-03/2026 **PhD (Tech)** *Vienna, Austria*

Institute of Telecommunications, TU Wien

Thesis: Millimeter-Wave Communications Using Sub-6 GHz Out-of-Band Information

10/2018-07/2021 **Master's Degree** *Vienna, Austria*

Institute of Telecommunications, TU Wien

Master's Programme Telecommunications

Thesis: Emulation Techniques for High Speed Train Measurements in 6G Mobile Communications

10/2013-07/2017 **Bachelor's Degree** *Sarajevo, Bosnia and Herzegovina*

Faculty of Electrical Engineering, University of Sarajevo

Bachelor's Programme Telecommunications

Thesis: Measurement of Error Probability in a Hardware-in-the-Loop Simulation Environment

Publications

Conference Paper
2026

F. Pasic, M. Mussbah, S. Schwarz and M. Rupp, **Beam-Domain Channel Estimation for mmWave MIMO using Sub-6 GHz Out-of-Band Information,** in Proceedings of the 20th International Symposium on Wireless Communication Systems (ISWCS), August 2026.

Conference Paper
2026

F. Pasic, M. Mussbah, S. Schwarz, M. Rupp, C. F. Mecklenbräuker, **Exploiting Out-of-Band Information for Millimeter-Wave MIMO Channel Estimation: Performance in Static and Dynamic Scenarios,** in Proceedings of the Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit), June 2026.

Conference Paper
2026

F. Pasic, J. Soklič, R. Langwieser, S. Schwarz, C. F. Mecklenbräuker, **Multi-Band Patch Antenna Array for Out-of-Band Aided Millimeter Wave Communication,** in Proceedings of the IEEE International Workshop on Antenna Technology (iWAT), March 2026.

Conference Paper
2025

F. Pasic, M. Mussbah, S. Schwarz, M. Rupp, F. Tufvesson, C. F. Mecklenbräuker, **Performance analysis of digital beamforming mmWave MIMO with low-resolution DACs/ADCs,** in Proceedings of the IEEE Radio and Antenna Days of the Indian Ocean (RADIO), October 2025.

Journal Paper
2025

F. Pasic, M. Mussbah, M. Hofer, S. Caban, S. Schwarz, T. Zemen, M. Rupp, C. F. Mecklenbräuker, **From sub-6 GHz to millimeter-wave: dynamic indoor measurements, channel characteristics and performance evaluation,** published in the IEEE Open Journal of the Communications Society, September 2025.

Journal Paper
2025

F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch, C. F. Mecklenbräuker, **Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information,** published in the IEEE Transactions on Communications, July 2025.

Journal Paper
2025

H. Hammoud, Y. Zhang, Z. Cheng, S. Sangodoyin, M. Hofer, F. Pasic, T. Pohl, R. Závorka, A. Prokeš, T. Zemen, C. F. Mecklenbräuker, A. F. Molisch, **Double-directional V2V channel measurement using ReRoMA at 60 GHz,** published in the IEEE Transaction on Vehicular Technology, July 2025.

Conference Paper
2025

F. Pasic, L. Eller, S. Schwarz, M. Rupp, C. F. Mecklenbräuker, **Deep learning-based mmWave MIMO channel estimation using sub-6 GHz channel information: CNN and UNet approaches,** in Proceedings of the IEEE INFOCOM 2025 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), May 2025.

Conference Paper
2025

M. Hofer, F. Pasic, B. Rainer, J. Blumenstein, A. Prokeš, C. F. Mecklenbräuker, A. F. Molisch, T. Zemen, **Enabling vehicular mmWave communication links using cmWave information,** in Proceedings of the 19th European Conference on Antennas and Propagation (EuCAP 2025), Mar. 30 - Apr. 4, 2025.

Conference Paper
2025

F. Pasic, M. Hofer, T. Zemen, A. F. Molisch, C. F. Mecklenbräuker, **Time-varying Rician K-factor in measured vehicular channels at cmWave and mmWave bands,** in Proceedings of the 19th European Conference on Antennas and Propagation (EuCAP 2025), Mar. 30 - Apr. 4, 2025.

Journal Paper
2024

A. F. Molisch, C. F. Mecklenbräuker, T. Zemen, A. Prokeš, M. Hofer, F. Pasic, H. Hammoud, **Millimeter-wave V2X channel measurements in urban environments,** published in the Open Journal on Vehicular Technology, December 2024.

Journal Paper
2024

H. Hammoud, Y. Zhang, Z. Cheng, S. Sangodoyin, M. Hofer, F. Pasic, T. Pohl, A. Prokeš, T. Zemen, C. F. Mecklenbräuker, A. F. Molisch, **A novel low-cost channel sounder for double-directionally resolved measurements in the mmWave band,** published in IEEE Transactions on Wireless Communications, Nov. 2024.

Conference Paper
2024

A. L. Gratzer, F. Schlägl, A. Schirrer, F. Pasic, M. Kolisnyk, C. F. Mecklenbräuker, S. Jakubek, **Modeling and analysis of human driver's compliance behavior to maneuver recommendations in form of soft inputs,** in Proceedings of the IEEE 27th International Conference on Intelligent Transportation Systems (ITSC), September 2024.

Conference Paper
2024

M. Hofer, D. Löschenbrand, F. Pasic, D. Radovic, B. Rainer, J. Blumenstein, C. F. Mecklenbräuker, S. Sangodoyin, H. Hammoud, G. Matz, A. F. Molisch, T. Zemen, **Similarity of wireless multiband propagation in urban vehicular-to-infrastructure scenarios,** in Proceedings of the IEEE 35th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2024), Sep. 2-5, 2024.

Conference Paper
2024

H. Hammoud, Y. Zhang, Z. Cheng, S. Sangodoyin, M. Hofer, F. Pasic, T. Pohl, A. Prokeš, T. Zemen, C. F. Mecklenbräuker, A. F. Molisch, **A novel low-cost channel sounder for double-directionally resolved measurements in the mmWave band,** in Proceedings of the IEEE International Conference on Communications (ICC 2024), Jun. 9-13, 2024.

Conference Paper
2024

F. Pasic, M. Hofer, M. Mussbah, S. Caban, S. Schwarz, T. Zemen, C. F. Mecklenbräuker, **Channel estimation for mmWave MIMO using sub-6 GHz out-of-band information,** in Proceedings of the Smart Applications, Communications and Networking (SmartNets 2024), May. 28-30, 2024.

Conference Paper
2024

D. Radovic, F. Pasic, M. Hofer, T. Zemen, C. F. Mecklenbräuker, **Stationarity of multiband channels for OTFS-based intelligent transportation systems,** in Proceedings of the 18th European Conference on Antennas and Propagation (EuCAP 2024), Mar. 17-22, 2024.

Magazine Paper
2023

D. Radovic, M. Hofer, F. Pasic, E. M. Vitucci, A. Fedorov, T. Zemen, **Methodologies for future vehicular digital twins,** submitted to IEEE Intelligent Transportation Systems Magazine, Oct. 2023.

Conference Paper
2023

F. Pasic, M. Hofer, D. Radovic, H. Groll, S. Caban, T. Zemen, C. F. Mecklenbräuker, **Quantifying the reproducibility of multi-band high speed wireless channel measurements,** in Proceedings of the IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2023), Sep. 5-8, 2023.

Conference Paper
2023

D. Radovic, F. Pasic, M. Hofer, T. Zemen, C. F. Mecklenbräuker, **Stationarity evaluation of high-mobility sub-6 GHz and mmWave non-WSSUS channels,** in Proceedings of the 2023 XXXVth General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS 2023), Aug. 19-26, 2023.

Magazine Paper
2023

F. Pasic, N. Di Cicco, M. Skocaj, M. Tornatore, S. Schwarz, C. F. Mecklenbräuker, V. Degli-Esposti, **Multi-band high-mobility measurements for deep learning-based channel prediction and simulation,** published in IEEE Communications Magazine (Volume: 61, Issue: 9, September 2023).

Conference Paper
2023

F. Pasic, M. Hofer, M. Mussbah, H. Groll, T. Zemen, S. Schwarz, C. F. Mecklenbräuker, **Statistical evaluation of delay and Doppler spreads in sub-6 GHz and mmWave vehicular channels,** in Proceedings of the IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Jun. 20-23, 2023.

Conference Paper
2022

F. Pasic, S. Pratschner, M. Rupp, C. F. Mecklenbräuker, **Pilot-aided channel estimation scheme for NR-V2X speed emulation technique,** in Proceedings of the 56th Annual Asilomar Conference on Signals, Systems, and Computers, Oct. 30 – Nov. 2, 2022.

Conference Paper
2022

F. Pasic, S. Pratschner, R. Langwieser, C. F. Mecklenbräuker, **High-mobility wireless channel measurements at 5.9 GHz in an urban environment,** in Proceedings of the International Balkan Conference on Communications and Networking (BalkanCom 2022), Aug. 22-24, 2022.

Conference Paper
2022

A. L. Gratzer, A. Schirrer, E. Thonhofer, F. Pasic, S. Jakubek, C. F. Mecklenbräuker, **Short-term collision estimation by stochastic predictions in multi-agent intersection traffic,** in Proceedings of the International Conference on Electrical, Computer and Energy Technologies (ICECET 2022), Jul. 20 – 22, 2022.

Conference Paper
2022

F. Pasic, D. Schützenhöfer, E. Jirousek, R. Langwieser, H. Groll, S. Pratschner, S. Caban, S. Schwarz, M. Rupp, **Comparison of sub 6 GHz and mmWave wireless channel measurements at high speeds,** in Proceedings of the 16th European Conference on Antennas and Propagation (EuCAP 2022), Mar. 27 – Apr. 1, 2022.

Conference Paper
2021

F. Pasic, S. Pratschner, R. Langwieser, D. Schützenhöfer, E. Jirousek, H. Groll, S. Caban, M. Rupp, **Sub 6 GHz versus mmWave measurements in a controlled high-mobility environment,** in Proceedings of the 25th International ITG Workshop on Smart Antennas (WSA 2021), Nov. 8-10, 2021.

Conference Paper
2019

F. Pasic, A. Maric, **RTL-SDR Based hardware-in-the-loop bit error rate measuring system,** in Proceedings of the 18th IEEE International Symposium INFOTEH-JAHORINA (INFOTEH), Jahorina, Bosnia and Herzegovina, Mar. 20-22, 2019.